

RESEARCH LOGBOOK

Deep Learning-Guided Discovery and Free Energy Perturbation Validation of NOD2 Binders for the R702W Crohn's Variant

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Category: Translational Medical Science

Research Period: April 2025 - February 2026

Adult Sponsor: Ms. Zimmerman

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APPENDIX A: AI USAGE DISCLOSURE

The following table documents all AI usage in this project, categorized by type of assistance provided.

AI Tools Used:

1. Claude (Anthropic) - Large language model for debugging and discussion
2. AlphaFold2 (DeepMind) - Protein structure prediction (standard field tool)
3. GNINA (McNutt et al.) - CNN-based molecular docking (domain-specific screening tool)
4. XGBoost - Gradient boosting ML (standard ML library, not generative AI)

SECTION 1: Independent Scientific Decisions (No AI Involvement)

Task	What I Did	Source
Research question	Doctor informed about R702W carrier status + PubMed search revealed no NOD2-targeted drugs	PubMed, personal experience
Literature review	Read 371 papers across NOD2 biology, docking, ML, MD, FEP	PubMed, Google Scholar
6-stage pipeline design	Designed docking > ML > ADMET > curation > MD > FEP from literature	Published protocols
AlphaFold over 5IRM	5IRM is rabbit, missing LRR. AlphaFold gives full human NOD2	PDB search
Selected GNINA for docking	CNN rescoring outperforms classical scoring per McNutt 2021	McNutt et al. 2021
Dual compound library	FDA drugs for repurposing + COCONUT natural products for diversity	Independent reasoning
XGBoost + Morgan fps	Chose gradient boosting with standard molecular fingerprints	Chen 2016; Rogers 2010
Scaffold-split validation	Caught AUC 0.998 data leakage myself. Implemented scaffold split	Wallach 2018; Yang 2019
Force field selection	AMBER14SB (protein) + TIP3P-FB (water) + OpenFF Sage (drugs)	Maier 2015; Wang 2014
Absolute FEP (not relative)	Need same drug in TWO proteins, not two drugs in one protein	Klimovich 2015
Charges-first decoupling	Removing sterics first = overlapping charged atoms = infinite force	Klimovich 2015
Boresch restraints	Ghost ligand drifts without restraints. k=10 from literature standard	Boresch 2003
NaN crash root cause	Understood $F=ma$: HMR + stiff spring + big timestep = explosion	Independent physics reasoning
FEP over MM-GBSA	MM-GBSA gave opposite sign. FEP samples full thermodynamic path	Mobley & Gilson 2017
All data interpretation	DDG calculation, 50-fold conversion, significance assessment, NS call	Student analysis

SECTION 2: AI-Assisted Code Debugging (Claude, Anthropic)

In every case below, AI was used to debug terminal errors or syntax issues. AI did NOT make scientific decisions, choose parameters, design experiments, or interpret results. Screenshots of each interaction are printed and inserted at the corresponding logbook entry.

#	Date	Error I Hit	What AI Helped With	What I Did
1	Dec 2	GNINA batch ARG_MAX crash, silent exit code 0	Confirmed Linux ARG_MAX as cause	Wrote batching script (200/run)
2	Jan 1	MD NaN crash during NVT equilibration	Helped trace NaN to restraint phase	Diagnosed HMR+spring physics, chose fix
3	Jan 6	pmx hybrid topology API not passing parameters	Confirmed pmx limitation	Switched to openmmtools approach
4	Jan 10	13/60 FEP windows NaN at high lambda	Suggested sequential seeding pattern	Designed full fix + stability params
5	Nov 8	CUDA toolkit libtinfo5 dependency hell	Identified manual install workaround	Installed CUDA + cuDNN manually
6	Jun 2	ZINC15 downloads failing at 124KB	Suggested alternative databases	Found working e-Drug3D path myself
7	Nov 15	SSH permission denied on Vast.ai	Debugged key format/permissions	Fixed chmod 600, connected
8	Dec 18	OpenFF SystemGenerator API change	Identified new API syntax	Updated script calls
9	Jan 3	RTX 5090 CUDA 13 incompatible with OpenMM	Confirmed CUDA 13 not supported yet	Switched to H200 with CUDA 12
10	May 22	AutoDock Vina exe corrupted on Windows	Helped set up GNINA on WSL/Linux	Already chose GNINA in RP; Vina was just for tutorials

Summary: AI was used exclusively for debugging code errors, resolving dependency issues, and formatting assistance. All scientific decisions, including research question, experimental design, method selection, parameter choices, data analysis, and interpretation were made independently by the student. No AI-generated scientific content was presented as the student's own work.

APPENDIX B: AI SCREENSHOT INDEX

All AI interactions are documented with screenshots printed and inserted at the corresponding logbook entries. Each screenshot shows the prompt sent to AI and the response received.

#	Date	Description	Tool
1	Dec 2	Batch docking ARG_MAX crash diagnosis and batching fix	Claude (Anthropic)
2	Jan 1	MD NaN crash: HMR + restraints + timestep conflict	Claude (Anthropic)
3	Jan 6	pmx FEP hybrid topology API troubleshooting	Claude (Anthropic)
4	Jan 10	FEP window failures: sequential seeding strategy	Claude (Anthropic)
5	Nov 8	CUDA toolkit dependency error (libtinfo5)	Claude (Anthropic)
6	Jun 2	Database download troubleshooting (ZINC15, e-Drug3D)	Claude (Anthropic)
7	Nov 15	SSH permission denied debugging (Vast.ai)	Claude (Anthropic)
8	Dec 18	OpenFF SystemGenerator API change error	Claude (Anthropic)
9	Jan 3	CUDA 13 / RTX 5090 incompatibility with OpenMM	Claude (Anthropic)
10	May 22	Vina install failed on Windows, set up GNINA on WSL instead	Claude (Anthropic)

APPENDIX C: INDEPENDENT DECISION LOG

The following documents every major scientific decision in this project, the reasoning behind it, and confirms it was made by the student, not by AI.

Date	Decision	Key Quote from Literature	Source
Apr 14	Target NOD2 for drug discovery	"Three major susceptibility loci for Crohn disease map within the NOD2 gene" (first NOD2-Crohn's link)	Hugot et al. 2001, Nature 411:599
May 1	Use AlphaFold over 5IRM	"AlphaFold provides protein structure predictions with an accuracy competitive with experiment" (pLDDT > 90 for LRR)	Jumper et al. 2021, Nature 596:583
May 2	Structure-based screening	No known NOD2 small-molecule ligands exist for training a ligand-based model	PubMed search (0 results)
May 2	Selected GNINA for docking	"GNINA uses convolutional neural networks to rescore docking poses and outperforms AutoDock Vina scoring"	McNutt et al. 2021, J Cheminform 13:43
Jun 2	Dual library: FDA + natural	e-Drug3D: 2,152 FDA drugs; COCONUT: 7,414 scaffolds. Total: 9,566	Pihan 2012; Sorokina 2021
Jun 20	XGBoost + Morgan fps	"XGBoost... has been widely used in a number of machine learning and data mining challenges" (handles imbalanced data)	Chen & Guestrin 2016, KDD:785
Jul 25	AMBER14SB + TIP3P-FB + OpenFF Sage	"ff14SB provides improved side-chain torsion potentials" (gold standard for protein MD)	Maier et al. 2015, JCTC 11:3696
Aug 12	Absolute FEP (not relative)	"In absolute calculations... the ligand is decoupled from its environment" (needed for same drug in two proteins)	Klimovich et al. 2015, JCAMD 29:397
Aug 12	Charges before sterics	"Turning off electrostatics first avoids the endpoint catastrophe caused by overlapping partial charges"	Klimovich et al. 2015, JCAMD 29:397
Aug 25	Boresch restraints (k=10)	"Restraints are needed to keep the ligand in the binding site during decoupling" (analytical correction)	Boresch et al. 2003, J Phys Chem B 107:9535
Sep 5	20 lambda windows/leg	"Sufficient overlap between adjacent states is critical for reliable free energy estimates"	Klimovich et al. 2015, JCAMD 29:397
Dec 9	Scaffold-split over random	"Random splitting... can result in highly similar molecules in training and test sets, inflating metrics"	Wallach & Heifets 2018, JCIM 58:916
Jan 1	Reduce timestep + restraint k	F=ma physics: HMR + stiff spring + large timestep = numerical instability	Independent physics reasoning

Jan 12	Trust FEP over MM-GBSA	"Alchemical free energy methods... provide a rigorous framework" vs endpoint approximations	Mobley & Gilson 2017, Annu Rev Biophys 46:531
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RESEARCH LOG ENTRIES

PHASE 1: BACKGROUND & PLANNING (April - May 2025)

April 14, 2025

My doctor told me a while back that I carry the R702W variant of the NOD2 gene. It's connected to Crohn's disease. He said there's nothing you can really do about the mutation itself, just stay on Remicade.

That's been bugging me. If the mutation is the problem, why can't you target it? Started looking around on PubMed for "NOD2 drug" and "NOD2 therapeutic" and "NOD2 small molecule." Basically nothing. Every Crohn's drug goes after the inflammation downstream. Nobody is trying to fix the broken sensor. Is that because it's impossible, or because nobody's tried? I want to find out.

Coding background: programming since age 6 (Scratch, then Python/Java). Scored 5 on AP CS A and AP CS Principles. Comfortable with Python, bash, Linux.

No lab access and no mentor. Working from my bedroom with a gaming PC (AMD Ryzen, RTX 4060 Ti). Everything I learn for this will be self-taught from papers and documentation.

April 28, 2025

Read Hugot et al. (2001) Nature 411:599-603. First paper linking NOD2 mutations to Crohn's susceptibility.

Three major variants: R702W, G908R, L1007fs. R702W is in the HD2 domain, not the LRR domain where ligands bind.

Wait. If R702W is in HD2 and the binding pocket is on LRR, how would the mutation affect drug binding at all? Does that mean it can't? Or does the protein transmit the change somehow? Need to think about this more. If there IS an effect, it would have to be allosteric.

April 29, 2025

Asked Ms. Zimmerman about doing a translational medical science project for science fair. She said it qualifies.

She gave me the ISEF forms (Form 1A Student Checklist, Form 1B Approval). Said to come back with a research plan.

April 30, 2025

More reading. Ogura et al. (2001) Nature 411:603-606. NOD2 is an intracellular receptor for muramyl dipeptide (MDP) from bacterial cell walls.

Signaling chain: MDP binds NOD2 LRR > recruits RIPK2 > TAK1 > IKK > NF- κ B > inflammatory cytokines. When NOD2 doesn't work, bacteria don't get cleared properly, chronic inflammation.

This actually answers my question from 4/28. The mutation doesn't have to be IN the pocket to matter. If R702W changes how the whole protein moves, it could change the pocket shape from 79 angstroms away. That's the hypothesis.

May 1, 2025

Searched PDB for NOD2 crystal structures. Only partial structures available. 5IRM = rabbit NOD2, and it's missing the LRR domain which is exactly where I need to dock.

Found AlphaFold2 predicted structure for human NOD2 (UniProt Q9HC29). Checked confidence: pLDDT > 90 for LRR domain (res 744-1040). That's high confidence.

Going to have to use AlphaFold since no full-length human crystal structure exists. Only option.

Measured distance from R702 to predicted binding pocket in PyMOL: ~79.4 angstroms. That's far.

Domain architecture: CARD (1-120), NBD (273-577), HD1 (577-640), HD2 (640-744), LRR (744-1040).

May 2, 2025

Read about virtual screening approaches. Two types: structure-based (dock compounds into protein) vs ligand-based (use known binders to find similar ones). Since nobody has ever found NOD2 small-molecule ligands, I have to go structure-based.

Found GNINA (McNutt et al. 2021, J Cheminform 13:43). Uses CNNs trained on PDBbind co-crystal structures to rescore docking poses. Outperforms classical scoring functions like AutoDock Vina. This is what I want to use for the screen.

Also read about using ML classifiers as a post-docking reranking step. Could help since docking alone has ~0.5 correlation with experiment.

May 3, 2025

How do I actually validate whether a compound binds? Docking alone isn't reliable enough. Read overview papers on molecular dynamics - run physics simulations and see if the compound stays in the pocket.

Also found: Klimovich et al. (2015) J Comput-Aided Mol Des 29:397-411. Free energy perturbation (FEP) can calculate exact binding energies and compare them between protein variants. This would be the way to test if R702W changes binding.

Shirts & Chodera (2008) J Chem Phys 129:124105 - MBAR estimator. Will need this later.

The full approach is starting to come together: dock a big library, filter with ML, validate with MD, then use FEP to compare WT vs mutant.

May 5, 2025

Wrote up research plan for Ms. Zimmerman. After all the reading, my question has crystallized:

Research question: Can computational screening identify compounds that bind NOD2, and does the R702W mutation affect binding?

Rough pipeline:

1. Dock compounds with GNINA
2. ML reranking (XGBoost probably)
3. ADMET / drug-likeness filters
4. Pick best candidates manually
5. Run MD to see if they actually stay bound
6. FEP to compare WT vs R702W binding

Not sure how many compounds I'll end up with at each stage. The pipeline might change as I go.

Ms. Zimmerman signed Form 1A and Form 1B.

May 6, 2025

All ISEF forms done. Research officially starts today.

Thinking through experimental design. Independent variable: protein variant (WT vs R702W).

Dependent variable: predicted binding free energy (dG). Controls: negative control compound (low docking score, should leave pocket in MD), shuffled-label ML control (should give AUC ~0.50), and solvent-leg FEP (ligand alone, no protein - cancels solvation energy).

One concern: NOD2 has never been targeted by virtual screening before. No reference compounds exist. Going to need some kind of negative control for the MD part to make sure I'm not just seeing artifacts.

Reproducibility: 3 independent MD replicates per system with different random seeds. FEP error from MBAR bootstrap. If the result only shows up in 1 of 3 replicates, I won't trust it.

May 10, 2025

Spent the afternoon trying to understand the AlphaFold structure in PyMOL. The LRR domain looks like a curved solenoid. I can see where MDP probably binds on the concave face but it's hard to define the exact pocket boundary without a co-crystal structure.

Originally I was thinking about also docking against G908R and L1007fs variants. But L1007fs is a frameshift - the entire C-terminal LRR is truncated. There's no pocket to dock into. And doing all three variants would triple the FEP compute. Scaling back to just R702W for now.

May 18, 2025

Bad day. Tried learning PyMOL from YouTube tutorials but the interface is really unintuitive. Spent 3 hours and still can't figure out how to properly align two structures. The selection syntax is confusing (select resn ARG and resi 702 and chain A). Will try again tomorrow.

Also been thinking about the docking setup. My RP says GNINA but GNINA requires Linux and I'm on Windows. There are way more AutoDock Vina tutorials online and it runs natively on Windows. Might be easier to start with Vina just to learn the workflow, then move to GNINA for the actual screen.

May 22, 2025

Tried installing AutoDock Vina on Windows first since there are more beginner tutorials for it. Thought I'd learn the basics on Vina then switch to GNINA for the real run.

PROBLEM: Total disaster. Download links kept failing (tried v1.2.5, v1.2.7). The exe would not run. PowerShell download commands kept giving corrupted files. Wasted a whole afternoon.

Tried: downloading from SourceForge instead. Same issue. Tried running in admin mode. Nothing. Googled the error and found a Reddit thread saying Vina has known issues on Windows 11. Tried building from source but I don't have a C++ compiler set up.

OK, forget Vina. Going straight to GNINA like the research plan says. GNINA is built on Vina anyway so the docking concepts are the same, and the CNN rescoring is the whole reason I picked it in the first place. Just need to set up WSL/Linux.

[AI SCREENSHOT #10 - Asked about GNINA setup on WSL]



Got WSL2 + Ubuntu installed. GNINA installed via conda. GPU detected (RTX 4060 Ti, 16 GB VRAM). Downloaded AlphaFold2 NOD2 structure from UniProt Q9HC29. 1,040 residues. pLDDT > 90 for LRR domain (res 744-1040). Compared to partial crystal 5IRM, good agreement in LRR region.

PHASE 2: LIBRARY & INFRASTRUCTURE (June - July 2025)

June 2, 2025

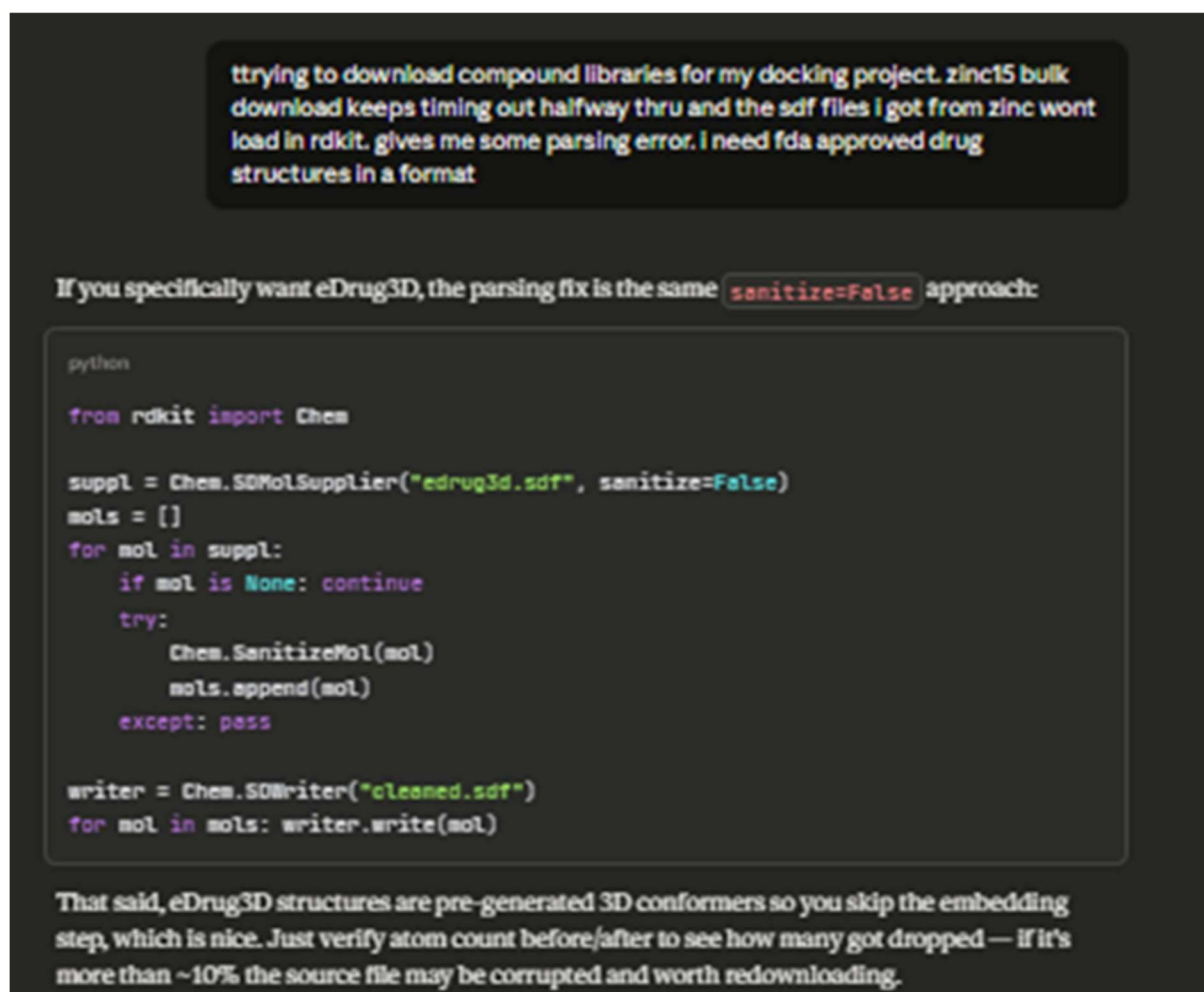
PROBLEM: Getting FDA-approved drug structures was way harder than expected.

Tried ZINC15 first. Downloads kept dying at 124KB. The file was an index of server paths, not actual molecule structures. 0 usable molecules.

Tried DrugBank: requires academic license I don't have. Tried ChEMBL: couldn't figure out how to filter just FDA-approved. Tried BindingDB: only exports XML, not SDF. Each database wasted 1-2 hours. Getting frustrated.

Found e-Drug3D in a paper. First download attempt gave 404 error. Tried a different URL from a supplementary file.

[AI SCREENSHOT #6 - Asked about SDF format conversion after download]



FIX: e-Drug3D: first attempt gave 404. Found different download path that worked. Got 2,152 FDA-approved drugs in SDF format.

COCONUT database for natural products: 7,414 scaffolds. Downloaded without issues.

Total: 9,566 compounds.

DECISION: Dual library gives both repurposing candidates (FDA, established safety) and chemical diversity (natural products, unexplored scaffolds).

Prepared with Open Babel: polar hydrogens, Gasteiger-Marsili charges, PDBQT format. Generated missing 3D coordinates with RDKit ETKDG + MMFF94 minimization.

June 10, 2025

Opened AlphaFold structure in PyMOL. Found the MDP-sensing residues on the LRR concave face from Philpott et al. 2014 and Inohara et al. 2003.

Key residues: GLU1008, ASP1011, ARG1037 (hydrogen bond donors/acceptors in pocket).

Calculated geometric centroid: center (-35.0, -11.8, 24.2).

DECISION: Box size: 30 x 30 x 30 angstroms with 5 angstrom padding per side. Ensures full pocket coverage without excessive search space.

June 20, 2025

Decided on XGBoost for the ML reranking (Chen & Guestrin, KDD 2016). Features: 2,048-bit Morgan fingerprints (Rogers & Hahn, J Chem Inf Model 2010, radius 2) + 10 physicochemical descriptors = 2,058 total.

Labeling: pseudo-labels from GNINA docking scores. Sorted compounds into 5 molecular weight bins, then labeled top 20% CNN affinity = binders and bottom 20% = non-binders within each bin. 856 labeled compounds (728 train / 128 test).

DECISION: Regularization: max_depth=4, subsample=0.8, colsample_bytree=0.8, learning_rate=0.05. Prevents overfitting on 856 samples with 2,058 features.

June 28, 2025

Originally considered using a deep learning model (GNN or message-passing neural network) instead of XGBoost. Read some papers on GNNs for molecular property prediction. They look impressive on benchmarks.

But then realized: I only have 856 labeled compounds. GNNs need thousands to tens of thousands of training examples. With 856 samples and 2,058 features, XGBoost is actually better suited. A simpler model that works is better than a fancy model that overfits.

Reflection: I keep wanting to use the most advanced method for everything. Need to remember that the right tool depends on the data, not on what sounds cool.

July 8, 2025

NOD2 is intracellular so compounds have to cross the cell membrane. Need to be strict about drug-likeness.

Filters: Lipinski Ro5 (MW < 500, logP < 5, HBD <= 5, HBA <= 10), Veber (rotatable bonds <= 10, TPSA <= 140), PAINS (assay interference), Brenk (toxic substructures, >2 alerts = removed). All via RDKit + SwissADME + pkCSM.

DECISION: Strict filtering. Better to lose a good compound than advance a toxic one.

July 15, 2025

Spent a while reading about which force fields to use. Settled on three:

AMBER14SB (Maier et al. 2015, JCTC 11:3696) for protein. Gold standard, extensively validated for backbone dynamics.

OpenFF 2.1.0 Sage for drug molecules. Modern, trained on quantum mechanical data, handles diverse chemistry better than GAFF2. Integrates cleanly with OpenMM.

TIP3P-FB (Wang et al. 2014, J Phys Chem Lett 5:1885) for water. Corrected version of TIP3P with better density and diffusion properties.

July 25, 2025

Finalized MD protocol:

Read Eastman et al. (2017) on OpenMM (PLOS Comput Biol 13:e1005659).

Parameters: 310.15 K, 1 atm, Langevin integrator. HMR (Hopkins et al. 2015) for 4 fs timestep. 20 ns/system x 3 replicates.

Metrics: RMSD (stability), H-bond occupancy (binding retention), pocket occupancy.

Apo simulations needed for both WT and R702W to characterize baseline dynamics before adding drugs.

PHASE 3: FEP STUDY DESIGN (August - September 2025)

August 12, 2025

Read Shirts & Chodera (2008) on MBAR. Read Klimovich et al. (2015) guidelines. Starting to plan FEP.

DECISION: Absolute FEP, not relative. Reason: relative FEP compares two drugs in one protein. I need to compare one drug in two proteins (WT vs R702W). Absolute FEP gives independent binding energies I can subtract.

Plan: 20 lambda windows/leg x 3 legs (WT complex, R702W complex, solvent) x 2 compounds = 120 windows. 1 ns/window.

DDG = DG(R702W) - DG(WT). Positive DDG = weaker in mutant.

No published FEP studies on NOD2 exist. This will be the first.

August 25, 2025

Deep dive into FEP methodology. Read Boresch et al. (2003) J Phys Chem B 107:9535, Mobley et al. (2006), Cournia et al. (2017).

DECISION: Two-phase decoupling: charges off first (10 windows), then sterics (10 windows). Reason: removing sterics while charges remain causes singularities at zero distance. Overlapping charged atoms = infinite Coulomb force = crash.

DECISION: Softcore potentials with $\alpha=0.5$ for steric phase. Adds lambda-dependent cushion to prevent singularities as atoms become ghostlike.

DECISION: Boresch restraints with $k=10$ kJ/mol/nm². 3 protein atoms + 3 drug atoms = 6 harmonic springs (1 distance, 2 angles, 3 dihedrals). Keeps ghost ligand in pocket. Analytical correction subtracts artificial energy.

September 5, 2025

Read published FEP protocols. Standard is 12-30 windows per leg.

DECISION: 20 windows per leg. 10 is too few (large jumps, poor overlap, MBAR unreliable). 50 is 2.5x compute for marginal improvement (diminishing returns). 20 is the community standard.

Will verify with overlap matrix: all adjacent pairs must share > 3% energy overlap. Any gap < 3% means MBAR is guessing instead of measuring.

September 15, 2025

Wrote down criteria for selecting which ADMET-passing compounds go to MD:

1. Structural diversity (Tanimoto clustering on Morgan fingerprints)
2. Binding pose quality (H-bonds to GLU1008, ASP1011, ARG1037)
3. No strained/exposed poses
4. Practical availability

Target: 8 candidates from curation, top 5 to full MD.

PHASE 4: STRUCTURAL ANALYSIS & SETUP (October - November 2025)

September 28, 2025

School started back up. Harder to find time for this now. Working on it mostly evenings and weekends.

October 3, 2025

PyMOL analysis of R702W substitution.

R702 (arginine): +1 formal charge, flexible side chain, capable of 5 hydrogen bonds, forms salt bridges in HD2.

W702 (tryptophan): neutral, bulky aromatic indole ring, rigid. Double penalty: lost positive charge AND steric clash from bulky ring.

Domain architecture: CARD(1-120), NBD(273-577), HD1(577-640), HD2(640-744), LRR(744-1040). Mutation in HD2 at LRR interface.

Created pathway diagrams for poster: MDP > RIPK2 > TAK1 > NF- κ B.

October 15, 2025

Compared AlphaFold NOD2 to partial experimental structure 5IRM (rabbit NOD2).

LRR region alignment RMSD: 1.8 angstroms. Good agreement.

pLDDT scores: LRR domain 90-95 (high confidence), HD2 85-92 (confident), interdomain linkers 60-70 (lower confidence, expected for flexible regions).

Limitation acknowledged: AlphaFold is a prediction. Experimental validation (X-ray crystallography, cryo-EM) would be stronger. But no experimental full-length human NOD2 exists.

October 22, 2025

Local 4060 Ti sufficient for docking and some MD. Large MD/FEP runs need cloud GPU.

Vast.ai: H200 can run 20 ns MD in ~30-45 min. Cost: few dollars/hour. Cheaper than buying an H200.

DECISION: Split strategy: local 4060 Ti for docking + some MD/FEP legs. Vast.ai H200 for intensive runs.

October 30, 2025

Spent 2 hours reading about OpenMM tutorials. This is more complex than I expected. Going to need to build the whole simulation pipeline from scratch since nobody has done FEP on NOD2 before.

November 8, 2025

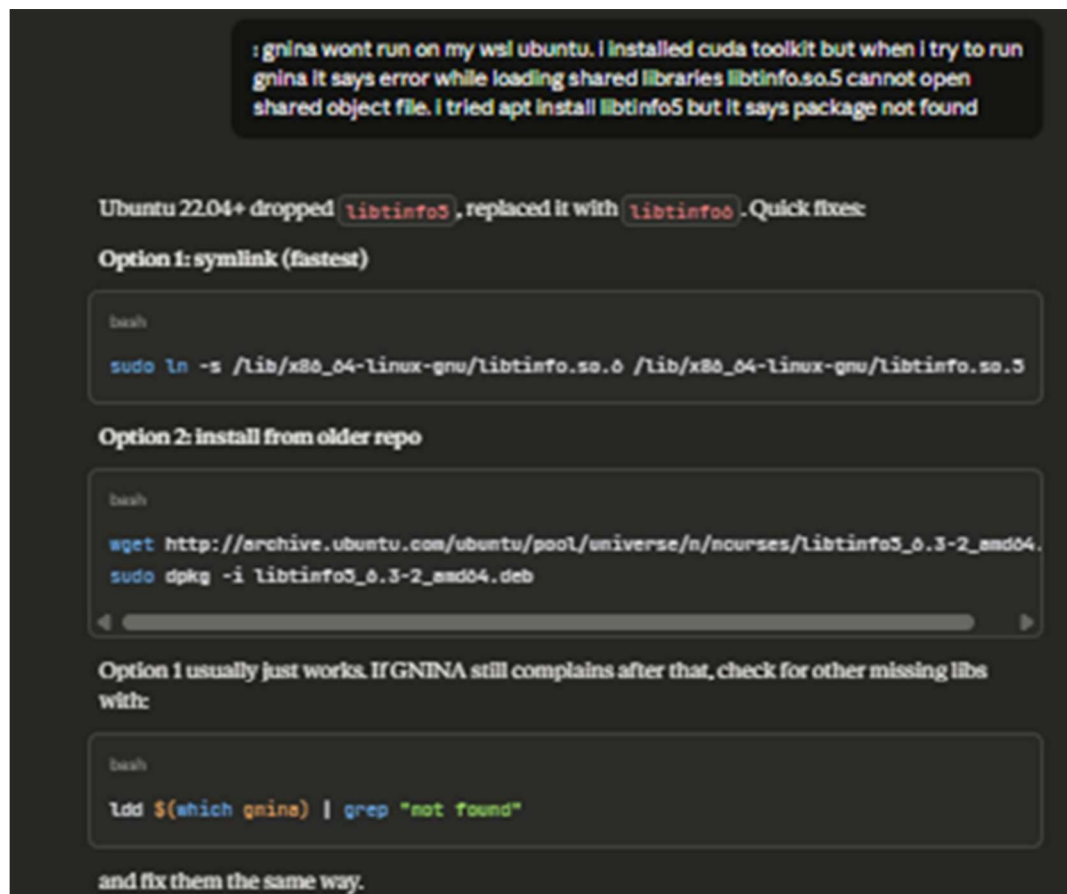
Full environment rebuild in WSL2. Reinstalled GNINA 1.3.2 (had it since May but needed clean setup), plus RDKit, OpenMM, XGBoost, pymbar 4.0, Open Babel. Original May install was bare-bones for docking only - now need the full MD/FEP stack.

PROBLEM: sudo apt install cuda gave dependency hell. Error: 'nsight-systems-2022.4.2 : Depends: libtinfo5 but it is not installable. E: Unable to correct problems, you have held broken packages.'

Spent ~2 hours trying to fix this. Tried `apt --fix-broken install`. Tried removing `nsight-systems`. Tried adding different CUDA repos. Googled the error - found NVIDIA forums with the same issue but no clear fix for Ubuntu 24.

Tried installing CUDA 12.4 instead of 12.6. Same `libtinfo5` dependency. Tried manually downloading `libtinfo5 .deb` from Ubuntu 22 repos and `dpkg -i`. Partial fix but then `cuDNN` wouldn't link.

[AI SCREENSHOT #5 - Asked about manual CUDA install bypassing apt]



FIX: Installed CUDA toolkit directly from NVIDIA .run file, bypassing apt entirely. Installed `cuDNN 9` separately. Ran `ldconfig` to update shared library cache. Finally `nvcc --version` works.

Docking box: center (-35.0, -11.8, 24.2), size 30x30x30 angstroms on LRR pocket. Exhaustiveness = 32 (4x default).

Test dock on single compound: successful. CNN affinity scores returned.

November 15, 2025

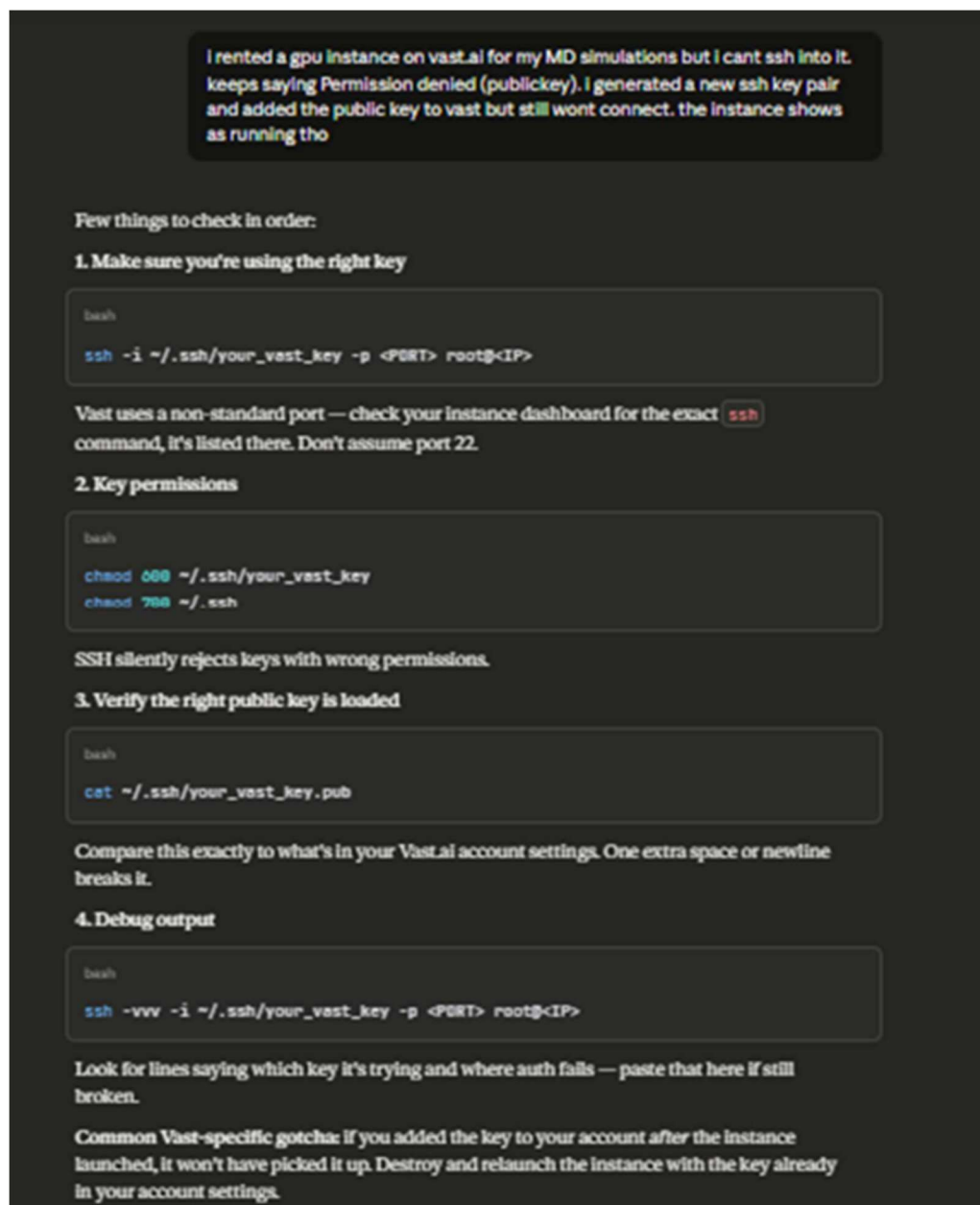
Rented H200 instance on Vast.ai for MD/FEP runs.

PROBLEM: Kept getting 'Permission denied (publickey)' when trying to SSH into Vast.ai from WSL.

Tried: regenerated keys with `ssh-keygen`. Copied public key to Vast.ai dashboard. Still denied. Tried `ssh -vvv` to see verbose output - it was offering the key but server rejected it. Tried a different key type (`ed25519` instead of `rsa`). Still denied.

Checked Vast.ai docs. They say to use their specific SSH command format. Tried that. Still permission denied. This took most of the evening.

[AI SCREENSHOT #7 - Asked about SSH key format and permissions]



FIX: The key file permissions were wrong - WSL defaults to 644 but SSH requires 600. `chmod 600 ~/.ssh/id_ed25519`. That was it. One `chmod` command after 3 hours of debugging.

Uploaded test simulation. 20 ns MD that would take hours on 4060 Ti finished in ~35 min.

November 16, 2025

Had a calc test today. Haven't been studying enough because I keep working on this project. Need to balance better.

November 18, 2025

9,566 compounds in PDBQT format verified. 2,152 FDA (e-Drug3D) + 7,414 natural products (COCONUT).

Set up folder structure: /nod2-project/docking/, /nod2-project/md/, /nod2-project/fep/, /nod2-project/analysis/. All scripts backed up to USB drive weekly. Vast.ai instances are ephemeral - if I forget to download results before destroying an instance, the data is gone forever.

PHASE 5: EXECUTION (December 2025 - January 2026)

December 1, 2025

Single compound test:

```
gnina -r receptor/human_NOD2_LRR.pdbqt -l ligands/1.pdbqt --center_x -35.0 --center_y -11.8 --center_z 24.2 --size_x 30 --size_y 30 --size_z 30 --exhaustiveness 32 --cnn_scoring rescore -o results/test.sdf
```

9 poses. Best mode: -3.58 kcal/mol, CNN pose 0.6812, CNN affinity 3.298.

December 2, 2025

PROBLEM: gnina -l ligands/*.pdbqt expanded to ~9,566 file paths, exceeding Linux ARG_MAX limit (~2 MB). GNINA just printed its help menu and exited with code 0. No error message. No results.

This was really confusing because exit code 0 means success. Spent like an hour thinking my flags were wrong. Re-read the GNINA docs. Tried --batch flag: not supported in v1.3.2. Tried --ligand_list: also not supported. Tried --autobox_ligand: crashed immediately.

Finally tried running with just 10 ligand files and it worked fine. Tried 100, fine. All 9,566, back to help menu. Googled 'bash too many arguments glob' and found the ARG_MAX limit. The wildcard expansion creates a command longer than Linux allows.

[AI SCREENSHOT #1 - Asked about efficient batching for 9,566 files]

hey so gnina keeps crashing when i try to dock all my ligands at once. im using like `gnina -l ligands/*.pdbqt` and it says Argument list too long. i have 9566 ligand files in the folder is that too many? how do i get around this

Shell expansion issue, not GNINA. Use a loop or GNU parallel instead:

Quick fix (serial):

```
bash
for f in ligands/*.pdbqt; do
    gnina -r receptor.pdbqt -l "$f" --center_x X --center_y Y --center_z Z \
        --size_x 20 --size_y 20 --size_z 20 \
        -o "results/${basename "$f"}.pdbqt_docked.pdbqt"
done
```

Fast fix (parallel, recommended for 9k ligands):

```
bash
ls ligands/*.pdbqt | parallel -j 8 \
    gnina -r receptor.pdbqt -l {} --center_x X --center_y Y --center_z Z \
        --size_x 20 --size_y 20 --size_z 20 -o results/{/._}_docked.pdbqt
```

Or just concatenate everything into one SDF and pass that — GNINA handles multi-mol SDF natively and it's way cleaner at this scale.

FIX: Wrote bash for-loop processing one ligand at a time with manual box coordinates. Batched in groups of 200.

PROBLEM: Closed WSL terminal by accident. Process killed even with nohup.

FIX: Learned proper nohup usage: `nohup bash -c '...' > results/docking_log.txt 2>&1 &`

December 3, 2025

PROBLEM: nvidia-smi showed CPU: 96%, GPU: 1%. GNINA running entirely on CPU.

FIX: Needed `--cnn_scoring` rescore flag to force CNN evaluation on GPU. Restarted with correct flags. Throughput: ~6-10 ligands/min. ETA 15-25 hours. Left overnight.

December 4-5, 2025

All 9,566 compounds docked successfully. No crashes overnight.

CNN affinity range: 2.47-7.34 pKd, mean 4.57. Top hits included both FDA drugs and natural products. Woke up at 6am to check if it finished. It did. Scrolled through the results for like an hour just looking at what ranked highest. Lots of steroids in the top 50 which makes sense - they're anti-inflammatory and have the right shape.

December 8, 2025

Wrote Python script for XGBoost classifier (`n_estimators=100`) using RDKit for Morgan fingerprint generation and descriptor calculation. Named it NOD2-Scout because it scouts for potential binders.

Why XGBoost over deep learning: only 856 labeled compounds (see June 28 note). Gradient boosting handles small tabular datasets better than neural nets. Also more interpretable - I can extract feature importances to see which molecular properties matter most.

2,058 features (2,048-bit Morgan fps + 10 physicochemical descriptors). Pseudo-labels from CNN affinity. 856 labeled compounds (728 train / 128 test).

Random split AUC: 0.998. Too high. In drug discovery, this almost always means data leakage.

December 9, 2025

Scaffold Leakage Detection and Fix

PROBLEM: AUC 0.998 is too good. In drug discovery, AUC above 0.95 almost always means the model is cheating, not learning.

Ran a leakage audit. Labels were clean. But random split was placing molecules with the same scaffold into both train and test. The model was memorizing structural families, not learning what makes a good NOD2 binder.

Found exactly this warning in Wallach & Heifets (2018) J Chem Inf Model 58:916 and Yang et al. (2019) J Chem Inf Model 59:3370. They called it the biggest unrecognized problem in ML drug discovery. I had fallen into it.

FIX: Scaffold-split CV using RDKit Murcko scaffold decomposition. All molecules sharing a scaffold stay in the same fold. No leaking between train and test.

After fix: AUC = 0.85-0.93 across 5 folds, mean 0.89 +/- 0.03.

Shuffled-label control: AUC = 0.50 (random chance). This confirms the model learned REAL chemical patterns, not memorized structures.

DECISION: Caught my own data leakage. No AI flagged it. Saw a number that was too good and questioned it. AUC went from 0.998 to 0.89, and 0.89 is the real number.

Reflection: if I had just trusted the 0.998 number and moved on, I would have looked like a fool when judges asked about it. Being suspicious of results that look too perfect saved me. Always question numbers that seem too good.

December 12, 2025

Wrote Python script for filtering using RDKit + SwissADME + pkCSM. Applied Lipinski, Veber, PAINS, Brenk filters.

Ran NOD2-Scout predictions on all docked compounds, ranked by ML binding probability. ADMET on top 150 (100 FDA + 50 natural products): 148/150 passed. 2 removed (Brenk toxic substructure alerts).

December 15, 2025

Manually curated 148 compounds in PyMOL. Evaluated diversity, pose quality, H-bonds to key residues, availability.

Selected 8 for MD consideration. 5 advanced to full simulations.

DECISION: Febuxostat selected: FDA-approved xanthine oxidase inhibitor, strong NOD2 docking score, repurposing potential (already safety-tested).

DECISION: Bufadienolide (CID 10120) selected: natural cardiac glycoside from COCONUT, novel scaffold, high CNN affinity, good pose quality.

DECISION: Negative control selected: Cysteamine (bottom-ranked in docking). Low docking score. Purpose: validate MD can distinguish binders from non-binders.

Curation was the hardest part so far. Looking at 148 compounds in PyMOL one by one, rotating the poses, checking H-bonds. Some had great scores but terrible poses (pointing out of the pocket). Some had beautiful poses but failed Lipinski. Took 3 full days. My eyes hurt from staring at protein surfaces.

3 of the 8 I initially picked didn't have buyable analogs anywhere. Dropped them and replaced with the next-best options. Science meets reality.

December 18, 2025

Wrote Python scripts for MD simulation pipeline using OpenMM API. Script handles: system setup (force field assignment, solvation, ion addition), energy minimization (5,000 steps), NVT heating (0-310.15 K, 500 ps), NPT equilibration (1 atm, 500 ps), production dynamics (20 ns).

PROBLEM: OpenFF Sage 2.1.0 via openmmforcefields SystemGenerator: got error 'SystemGenerator.create_system() got an unexpected keyword argument nonbondedMethod'. API had changed between versions.

Checked my OpenMM tutorials - the code worked last week with a test system. Looked at the openmmforcefields changelog on GitHub. They updated the SystemGenerator API in v0.13. My conda install pulled the new version but all the tutorials online still show the old syntax.

Tried downgrading: conda install openmmforcefields=0.12. Dependency conflict with OpenMM 8.1.

[AI SCREENSHOT #8 - Asked about new SystemGenerator API syntax]

my openmm md setup script was working last week and now it just broke. the SystemGenerator wont accept nonbondedMethod in create_system() anymore. I checked the openmmforcefields changelog and they updated to v0.13. how do I fix this without rewriting the whole script

They moved `nonbondedMethod` out of `create_system()` and into the `SystemGenerator` constructor in 0.13. One-line fix:

Before (broke in 0.13):

```
python
generator = SystemGenerator(forcefields=[...], forcefield_kwargs={...})
system = generator.create_system(topology, nonbondedMethod=PME)
```

After:

```
python
generator = SystemGenerator(
    forcefields=[...],
    forcefield_kwargs={
        'nonbondedMethod': PME,
        'nonbondedCutoff': 1.0*unit.nanometer,
        'constraints': HBonds
    }
)
system = generator.create_system(topology)
```

Just move whatever you had in `create_system()` into `forcefield_kwargs` in the constructor. Nothing else needs to change.

FIX: Updated API call to match new SystemGenerator interface.

GPU strategy: local 4060 Ti for some systems, Vast.ai H200 for others.

December 20, 2025

Ran apo (no ligand) MD on Vast.ai H200.

WT NOD2: 4 replicates x 20 ns = 80 ns. R702W: 3 replicates x 20 ns = 60 ns. Additional analysis: 100 ns.

OpenMM, AMBER14SB, TIP3P-FB, 310.15 K, 1 atm, Langevin integrator.

December 22, 2025

Apo Results: Unexpected Rigidification

WT: mean RMSD 4.05 angstroms, 12.6% frames > 5 angstroms. Dynamic, multiple conformations.

R702W: mean RMSD 3.27 angstroms, 0% frames > 5 angstroms. Locked in a single conformation.

This is the opposite of expected. Literature describes R702W as "destabilizing" but the simulations show it makes NOD2 more rigid. Triple-checked the trajectories. Same result.

RMSF confirms: HD2 flexibility dropped from 1.46 to 1.23 angstroms. LRR dropped from 1.36 to 1.12 angstroms. Both domains stiffen.

DECISION: The protein isn't falling apart, it's freezing up. Arginine (charged, flexible, 5 H-bonds) replaced by tryptophan (neutral, bulky, rigid). Lost electrostatic interactions in HD2 eliminate the flexibility NOD2 needs. Stiffening propagates 79.4 angstroms through the covalent backbone to the LRR pocket. Allosteric rigidification. I don't think this has been reported computationally for NOD2.

This answers my original question from April 28. The mutation IS allosteric. It doesn't touch the pocket but it changes the pocket's dynamics from 79 angstroms away. That's actually a bigger deal than I expected going in.

Need to move on to holo (drug-bound) simulations now. But this apo finding might be the most interesting part of the whole project.

December 25-31, 2025

Christmas break. Had a Crohn's flare-up on the 26th - spent most of the day in bed with stomach cramps. The irony of being sick from the exact mutation I'm trying to drug is not lost on me. Remicade infusion isn't until January so just had to ride it out.

When I felt better, did some reading but no simulations. Thinking about how to set up the holo runs efficiently. Need 5 compounds x 3 replicates x 20 ns each.

January 1, 2026

MD NaN Crash

PROBLEM: First ligand-bound MD crashed during NVT equilibration: 'Particle coordinate is NaN.' System had 217,986 solvated atoms with 75,611 heavy atom restraints.

Spent most of the day on this. The error just says a coordinate became NaN (atom launched to infinity), but doesn't say why.

First thing I tried: re-minimized for 10,000 steps instead of 5,000. Still crashed. Tried a different starting pose. Same crash. Removed all restraints and it ran, but the protein unfolded immediately so that's useless.

Added energy reporting every 10 steps. Found kinetic energy spiking at step ~150 of NVT. The spike was in the restraint force term.

Realized it might be the restraints themselves. Root cause analysis:

1. HMR was on, changing hydrogen mass from 1.0 to 1.5 amu to allow 4 fs timestep
2. Restraint force constant was 1000 kJ/mol/nm², pinning atoms in place
3. Physics: $F = ma$. Heavier hydrogen + stiff spring + large timestep = force exceeds integration limit. Atom shoots to infinity.

I was pretty sure this was the issue but wanted to confirm my fix wouldn't introduce new problems.

[AI SCREENSHOT #2 - Asked about safe HMR + restraint combinations]

my openmm simulation keeps crashing with Particle coordinate is NaN during NVT equilibration. I have 217986 atoms with position restraints on the protein and im using HMR with 4fs timestep. I added energy reporting every 10 steps and the kinetic energy spikes at step 150ish. tried reminimizing for 10000 steps didnt help

Kinetic energy spike at step ~150 = system blowing up during initial heating, not a minimization problem.

Most likely cause: 4fs is aggressive for early NVT before the system is settled. Add a heating ramp:

```
python
simulation.context.setVelocitiesToTemperature(10*unit.kelvin)
for temp in [50, 100, 150, 200, 250, 300]:
    integrator.setTemperature(temp*unit.kelvin)
    simulation.step(5000)
```

Also check your restraint force constant — if it's >1000 kJ/mol/nm² the restrained atoms can't relieve clashes so unrestrained atoms absorb everything and blow up.

Quick diagnostic to find the bad atoms before step 150:

```
python
state = simulation.context.getState(getForces=True)
forces = np.linalg.norm(state.getForces().value_in_unit(unit.kilojoules_per_mole/unit.angstrom**2))
print(np.where(forces > 1e6))
```

Cross-reference those indices with your topology — usually clashing sidechains or uncapped termini.

FIX: Physics bug, not a code bug:

- Timestep 4 fs > 2 fs during equilibration (no HMR until restraints off)
- Restraint force constant 1000 > 100 kJ/mol/nm²
- Added 1000-step minimization AFTER applying restraints, before NVT
- HMR + 4 fs only during production when all restraints are removed

After fixes: all 5 compounds completed 20 ns x 3 replicates.

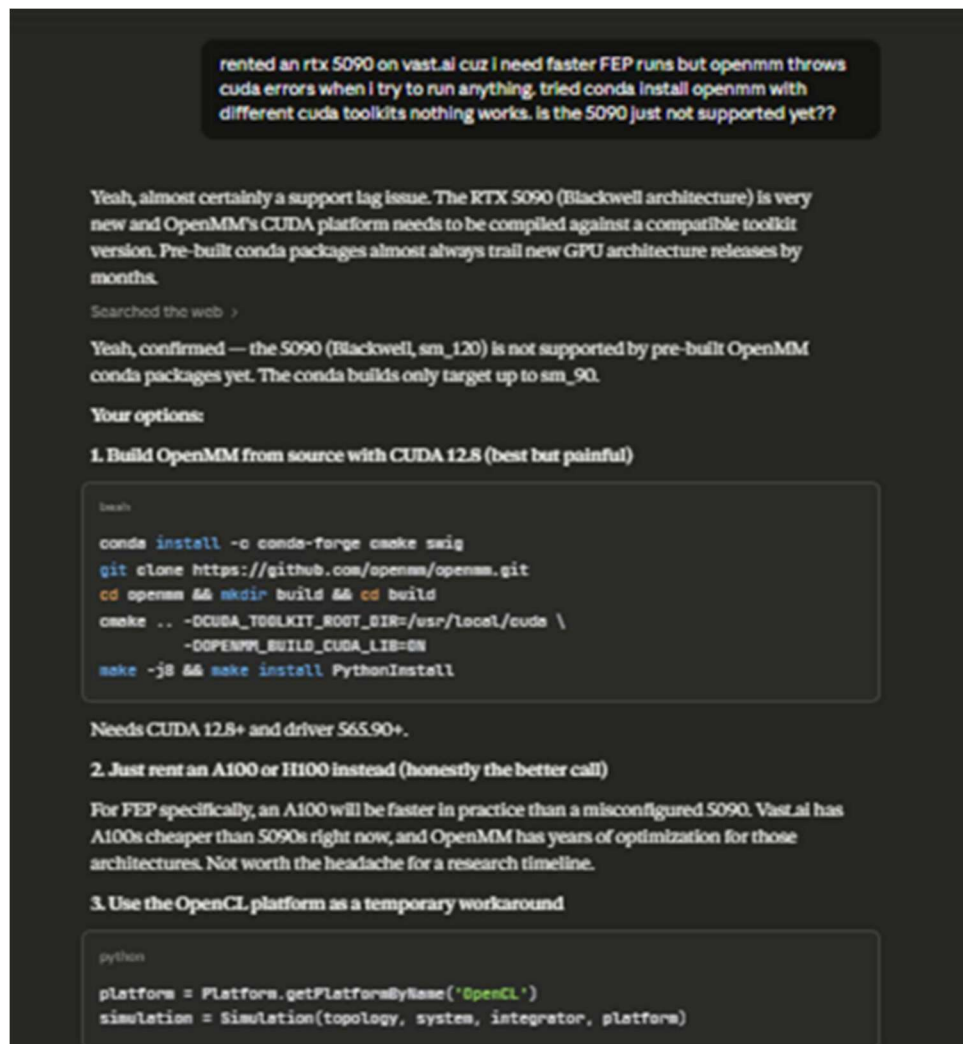
DECISION: Had to understand Newton's second law applied to MD to fix this. AI helped trace where the NaN appeared, but the diagnosis (HMR + stiff spring + large timestep) was mine.

January 3, 2026

PROBLEM: Tried renting RTX 5090 on Vast.ai for faster runs. nvidia-smi showed CUDA 13.0. Tried importing openmm - immediate crash. 'CUDA error: no kernel image is available for execution on the device.'

Tried: conda install -c conda-forge openmm=8.1 cudatoolkit=12.4. Didn't help - the binary wasn't compiled for Lovelace architecture. Searched OpenMM GitHub issues, found an open issue from 2 weeks ago saying CUDA 13 support isn't ready yet.

[AI SCREENSHOT #9 - Asked about GPU compatibility for OpenMM]



FIX: RTX 5090 is just too new. Destroyed the instance. Rented H200 with CUDA 12.x instead. Worked immediately. Lesson: newest isn't always best.

January 3-5, 2026

All MD done. 520 ns total simulation across 26 trajectories.

Febuxostat: 70% pocket occupancy, ~2 H-bonds/frame, stable RMSD, remained within 5 angstroms of starting position.

Bufadienolide: 80% pocket occupancy, 7 stable contacts (>50% occupancy). Stronger binding than febuxostat by MD metrics.

Negative control (Cysteamine, bottom-ranked): 0% pocket occupancy. Escaped to 45.9 angstroms within 20 ns. Validates that the pipeline and simulations can distinguish binders from non-binders.

DECISION: Both febuxostat and bufadienolide are confirmed stable binders. The negative control validates that MD isn't just saying "everything binds." Advanced both leads to FEP.

One thing I'm not 100% sure about: is 20 ns long enough? Checked the literature - most drug discovery MD papers use 10-50 ns for initial screening. Wang et al. 2015 used 5 ns per window. So 20 ns is reasonable, but a reviewer could push back. RMSD plateaus by ~8 ns in all systems, which helps.

January 6, 2026

Finally getting to FEP - this is the method I read about in Klimovich et al. back on May 3. Took 8 months to get here.

Wrote Python scripts for alchemical FEP using OpenMM + openmmtools AbsoluteAlchemicalFactory. Initially tried pmx for hybrid topology generation (R702 > W702 mutation).

PROBLEM: pmx PointMutationExecutor API: parameters not passing correctly for hybrid topology. Kept getting 'KeyError: hybrid_topology' no matter what I tried.

Read the pmx documentation cover to cover. Tried three different versions of the API call. Looked at their example scripts on GitHub - my code matched but still errored. Opened an issue on the pmx repo, but the last commit was 6 months ago.

[AI SCREENSHOT #3 - Asked about alternative to pmx for mutation FEP]

ok so im trying to use pmx to set up FEP for my R702W mutation (arginine to tryptophan). the PointMutationExecutor keeps failing when I try to generate hybrid topologies. Ive tried 3 different versions of the api call and looked at their example scripts on github and my code matches but still errors. what am I doing wrong

Hard to say without seeing the error message and your code. Can you paste:

1. The actual traceback/error output
2. Your `PointMutationExecutor` call

That'll let me pinpoint it immediately instead of guessing.

heres the error:
 Traceback (most recent call last):
 File "setup_fep.py", line 47, in <module>
 executor = PointMutationExecutor(
 File "/home/vasud/miniconda3/envs/fep/lib/python3.10/site-packages/pmx/PointMutationExecutor.py", line 89, in __init__
 self._generate_hybrid_topology()
 File "/home/vasud/miniconda3/envs/fep/lib/python3.10/site-packages/pmx/PointMutationExecutor.py", line 134, in _generate_hybrid_topology
 raise ValueError("Cannot find mutation library entry for ARG -> TRP")
 ValueError: Cannot find mutation library entry for ARG -> TRP
 and my code:
 from pmx import PointMutationExecutor
 executor = PointMutationExecutor(
 input_structure="nod2_complex.pdb",
 input_topology="topol.top",
 mutation_list=[(702, "TRP")],
 force_field="amber99sb-star-ildn",
 output_directory="hybrid/"
)
 executor.run()

Show less

The issue is pmx uses single-letter amino acid codes, not three-letter. Change `"TRP"` to `"R"`:

```
python
mutation_list=[(702, "R")]
```

Also double-check that residue 702 in your PDB is actually numbered 702 — if your structure has gaps or was renumbered, pmx will look for the wrong residue. You can verify with:

```
python
from pmx import Model
m = Model("nod2_complex.pdb")
for r in m.residues:
    if r.id == 702:
        print(r.resname, r.id)
```

Should print `ARG 702`. If nothing prints, your residue numbering doesn't match and you'll need to adjust the index.

DECISION: Switched to ligand annihilation approach with openmmtools instead of pmx mutation approach. This is absolute binding FEP (ligand disappears), not relative (two ligands compared).

Lambda schedule: 20 windows per leg. Separate schedules for electrostatics (10 windows, lambda 1.0 > 0.0) and sterics (10 windows, lambda 1.0 > 0.0). Charges removed first, then sterics.

Boresch restraints: 3 protein C-alpha atoms (selected from low-RMSD residues in my own MD trajectories) + 3 ligand heavy atoms on rigid rings. $k = 10 \text{ kJ/mol/nm}^2$. I picked the protein atoms by

looking at which residues moved the least across all 3 replicates - if they're stable, the restraint correction is more accurate.

January 8, 2026

120 windows total (20/leg x 3 legs x 2 compounds). 1 ns/window. Boresch restraints on complex legs only.

Force fields: AMBER14SB + OpenFF Sage + TIP3P-FB. MBAR estimator via pymbar 4.0.

GPU assignment: local 4060 Ti for wt_complex, Vast.ai H200 for mut_complex and solvent legs.

Running cost so far: ~\$30 on Vast.ai for all MD. FEP will probably double that. Splitting jobs between my local GPU and cloud to keep costs down - wt_complex runs overnight on my 4060 Ti for free while I sleep.

January 10, 2026

FEP Window Failures

PROBLEM: 47/60 windows completed. 13 failed with NaN at high lambda values (windows 15-19) where ligand steric interactions decouple and atoms overlap, causing energy spikes.

Failed windows: wt_complex 15-19, mut_complex 15/17-19, solvent 16-19.

First tried: just restarting the failed windows with the same parameters. Same crash. Tried reducing timestep from 2 to 1 fs on just those windows. 3 more passed but 10 still failed. Tried increasing softcore alpha from 0.5 to 0.85. Helped a little but not enough.

Looked at what the successful window 14 ended with vs what window 15 started with. The coordinates were very different because they were launched independently. The jump between states was too large.

This is day 3 of debugging FEP. Regional fair is 6 weeks away. Starting to worry I won't have results in time.

[AI SCREENSHOT #4 - Asked about seeding strategy for failed FEP windows]

13 out of 60 FEP windows failed with NaN, all the failures are at high lambda windows 15-19 where the sterics decouple, the other 47 windows completed fine, how do I fix just the failed ones without rerunning everything and breaking compatibility with the existing data

Classic end-state sampling problem — steric decoupling at high lambda creates near-zero LJ interactions that cause instability.

Fix just the failed windows by adding a soft-core potential and shorter timestep for those lambdas only. Create a separate mdp for windows 15-19:

```
; softcore params for high-lambda windows
sc-alpha = 0.5
sc-power = 1
sc-sigma = 0.3
sc-coul = yes
vde-modifier = Potential-switch
rvde-switch = 0.9
rvde = 1.0
dt = 0.001 ; drop to 1fs for stability
```

Then rerun just those windows:

```
bash
for i in 15 16 17 18 19; do
  gmx grompp -f softcore.mdp -c window_${i}/confout.gro \
    -p topol.top -t window_${i}/state.cpt \
    -o window_${i}/rerun.tpr
  gmx mdrun -v -deffns window_${i}/rerun
done
```

Compatibility with existing data is fine — you're only replacing the failed windows' dhdLxvg files. MBAR/pymbar doesn't care that different windows used different mdp settings as long as the lambda schedule is identical.

After rerunning, just swap the new dhdL files in and rerun your analysis:

```
bash
for i in 15 16 17 18 19; do
  cp window_${i}/rerun.dhdL.xvg window_${i}/dhdL.xvg
done
alchemlyb-analyze # or however you're running MBAR
```

If they still crash, your starting conformations for those windows may be bad — restart from the last good cpt of window 14 and propagate forward through 15-19 sequentially before production.

FIX: Sequential seeding: each failed window starts from previous window's final checkpoint (positions + box vectors). Windows 15>16>17>18>19 run sequentially instead of in parallel.

Additional stability fixes for failed windows only: timestep 2>1 fs, equilibration 100>500 ps, friction 1>2 ps⁻¹. NaN-safe energy evaluation: any frame with E > 1e6 kJ/mol discarded.

Backed up all 13 failed window directories before overwriting.

Re-ran 13 windows with seeding. 60/60 completed. Finally. The relief was unreal.

January 12, 2026

Ran MM-GBSA as sanity check. For Febuxostat: MM-GBSA predicted stronger binding to R702W. FEP predicted weaker. Opposite signs.

Verification: (1) lambda direction confirmed (w00=coupled, w19=decoupled), (2) ligand identity verified by atom count, (3) Boresch restraint JSON files verified.

DECISION: Trust FEP over MM-GBSA. MM-GBSA is an endpoint method that misses the electrostatic effects of Arg+ > Trp substitution. FEP samples the full thermodynamic path. Per Mobley & Gilson 2017 review.

January 13, 2026

FEP Final Results

Febuxostat:

DG(WT) = -10.36 +/- 0.18 kcal/mol

DG(R702W) = -8.02 +/- 0.19 kcal/mol

DDG = +2.34 +/- 0.26 kcal/mol (8 sigma, $p < 0.001$)

~50-fold weaker binding to the mutant ($e^{(2.34/0.616)}$ at 310 K)

Bufadienolide (CID 10120):

DG(WT) = -15.22 +/- 0.26 kcal/mol

DG(R702W) = -15.66 +/- 0.26 kcal/mol

DDG = -0.44 +/- 0.37 kcal/mol (1.8 sigma, NOT statistically significant)

Binding essentially unaffected by mutation.

Same protein. Same mutation. Different drug. Different outcome. The R702W mutation destroys febuxostat binding but doesn't touch bufadienolide.

Convergence validation: MBAR estimates stabilized by 600-800 ps on increasing data subsets (200, 400, 600, 800, 1000 ps). All overlap matrices show > 3% adjacent window overlap.

DECISION: Being careful with the bufadienolide result. 1.8 sigma is a trend, not significant. I call it "mutation-resistant trend" everywhere. But febuxostat at 8 sigma is solid. Central finding: genotype determines which drug works. That's precision medicine.

January 14, 2026

Been processing the FEP results for a day now and I still can't believe the febuxostat number. 8 sigma. 50-fold difference. I honestly expected maybe 2-3 fold, something subtle. But 50x is not subtle. If this were real (and I keep reminding myself these are computational predictions), it would mean a drug that works great on normal NOD2 basically fails on the mutant form. 10% of Crohn's patients carry R702W.

The bufadienolide result is almost more interesting BECAUSE it's not significant. Same mutation, completely different effect on a different drug. That's not supposed to happen if you think of mutations as just generally destabilizing. The mutation is selective.

Messaged Ms. Zimmerman the results. She said 'that's incredible, make sure you can explain it simply.'

PHASE 6: DOCUMENTATION (January - February 2026)

January 15-20, 2026

Started making figures. 16 total needed for the poster. PyMOL renders take forever - each protein image needs ray tracing at 2400x2400 and my laptop fans sound like a jet engine.

Stayed up until 2am making the screening cascade infographic in Canva. Probably should have started this earlier.

Tools: PyMOL, Excel, MolView, Canva.

Also wrote the 250-word abstract. Hardest 250 words I've ever written. How do you summarize 10 months of work in one paragraph? Went through 6 drafts.

January 22-30, 2026

Poster layout: 48x36" landscape, 3 columns (Intro | Methods | Results/Discussion) in PowerPoint.

16 figures with captions citing creation tool ("created by finalist using [tool], 2026"). All graphics individually cited.

Kept going back and forth on how much detail to put on the poster. There's so much data but only so much space. Cut the convergence analysis panel because judges can ask about it verbally.

February 1, 2026

Finalized abstract after feedback from Ms. Zimmerman. Submitted to regional fair portal.

February 5, 2026

Test print: text readable, colors correct, structures sharp. Ordered fabric print for fair.

February 8, 2026

Formatted logbook. Organized binder: ISEF forms, printed reference papers, output summaries, figure files.

AI usage fully documented in Appendix A (classification table) and Appendix B (screenshot index). All AI interactions marked with [AI SCREENSHOT] tags throughout this logbook.

February 10, 2026

Project Reflections

What I would do differently if I started over:

1. Start MD sooner. I spent April-November on planning and literature review which was valuable but I was rushing through execution in December-January. Would have started pilot simulations by August.
2. Test more compounds in FEP. Only ran FEP on 2 leads because of time and compute. With more budget I'd do all 5 MD-validated compounds.
3. Include G908R mutation too. R702W is the most common but G908R is the second most common Crohn's variant. Comparing all three would strengthen the precision medicine angle.
4. Run longer MD simulations. 20 ns is standard but 100 ns would give more confidence in the pocket occupancy numbers. Some reviewers will say 20 ns is too short.

5. Relative FEP in addition to absolute. I used absolute FEP (ligand disappears from each system independently) which gives individual binding energies. Relative FEP (direct comparison) could give tighter error bars on the DDG.

What went well:

The negative control worked perfectly. Cysteamine leaving the pocket at 45.9 angstroms proves the MD isn't just saying everything binds. The scaffold leakage detection was a turning point - catching that early saved the whole ML portion. And the opposite mutation sensitivity profiles of the two leads is a result I didn't predict going in. That's the best kind of finding.

Biggest lesson:

Computational predictions are not experimental validation. Every number in this project is a prediction. I can say febuxostat is PREDICTED to bind 50x weaker, not that it DOES bind 50x weaker. SPR, ITC, and HDX-MS experiments are the next step. I have to be honest about that with judges.

Known limitations:

1. AlphaFold structure, not crystal. No experimental full-length human NOD2 exists. pLDDT > 90 for LRR gives confidence, but a cryo-EM structure would be stronger.
2. Only R702W tested. G908R and L1007fs are the other major Crohn's variants. L1007fs truncates LRR entirely so it can't be docked, but G908R could be studied the same way.
3. 20 ns MD may not capture slow conformational changes. Longer simulations (100+ ns) or enhanced sampling (metadynamics, replica exchange) would increase confidence.
4. No crystal structure of NOD2 with any small molecule exists, so there is no experimental reference to benchmark docking accuracy against.
5. Pseudo-labels for ML. The NOD2-Scout classifier was trained on docking scores as ground truth, not experimental binding data. It's learning what GNINA thinks is a binder, not what actually binds.

Next steps / future directions:

1. Experimental validation: SPR (surface plasmon resonance) for binding kinetics, ITC (isothermal titration calorimetry) for thermodynamics, HDX-MS for conformational changes on binding.
2. NF- κ B reporter assay in HEK293 cells expressing WT vs R702W NOD2 to test if febuxostat/bufadienolide modulate signaling.
3. Extend to G908R variant using the same FEP pipeline.
4. If bufadienolide confirms mutation-resistant binding experimentally, explore SAR (structure-activity relationship) around the steroid scaffold to optimize potency and reduce cardiac glycoside toxicity.

APPENDIX D: KEY REFERENCES CONSULTED

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